

Data efficiency: accuracy versus number of training proteins

Experiment 03 · ddG_with_Boltz · raw-Δz features · gradient-boosted trees · protein-holdout.

Pooled accuracy is **near-saturated** in the number of training proteins: just **33 proteins** already reach Pearson $r = 0.74$, and a **10× increase** to 330 proteins adds only **+0.05** (to 0.79). The raw-Δz representation is strong enough that the predictor is not badly data-starved; adding proteins yields diminishing returns.

1. Motivation

Does accuracy scale with the number of distinct training proteins — i.e. is the predictor data-limited (more proteins would help) or saturated (the representation has extracted most of what it can)? And how few proteins are needed to reach useful performance?

2. Methods

Five-fold grouped cross-validation by protein (test proteins never appear in training). For each training fraction we subsample that fraction of the available training proteins, fit a gradient-boosted-tree regressor on 256 raw-Δz features, and pool out-of-fold predictions. Every fraction below 1.0 is averaged over three random protein subsamples; the run-to-run standard deviation is ≤ 0.002 , so the trend is robust.

3. Results

Fraction	~#train proteins	pooled Pearson r	RMSE	MAE
0.10	33	0.744	0.668	0.488
0.25	82	0.767	0.642	0.466
0.50	165	0.781	0.626	0.455
1.00	330	0.793	0.610	0.444

Table 1. Pooled metrics vs number of training proteins (test proteins held out).

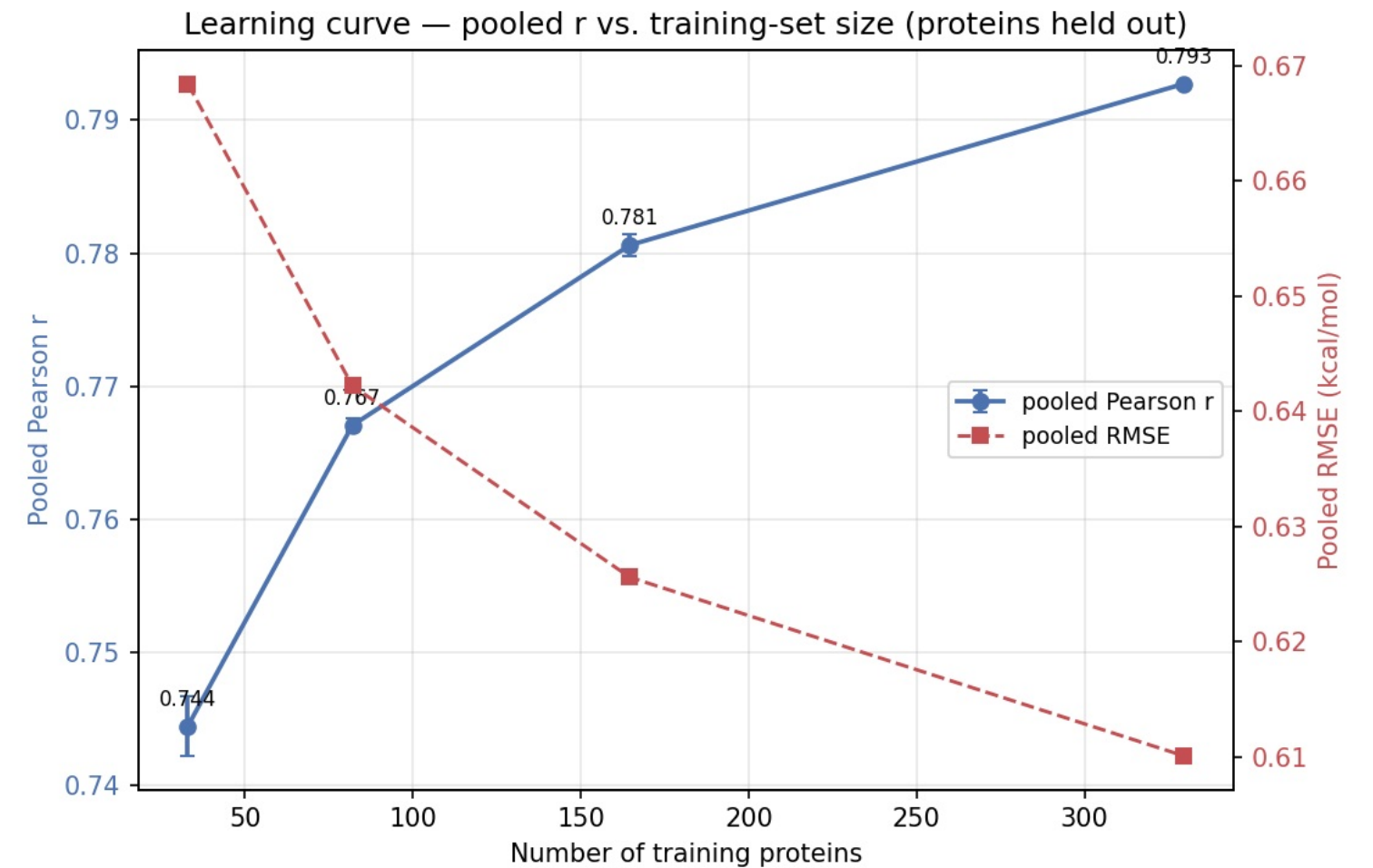


Figure 1. Pooled Pearson r (left axis) and RMSE (right axis) vs number of training proteins; shaded band is the seed SD.

4. Discussion

The curve is concave and flattening. Most of the achievable accuracy is reached with a few dozen proteins, and the marginal value of additional proteins falls quickly. This indicates that the raw-Δz features already capture most of the transferable stability signal, and that scaling the number of *proteins* is not the primary lever for further gains — consistent with the finding elsewhere that residual error is

governed by coverage of the $\Delta\Delta G$ range rather than by the raw count of training proteins.